

Sales Forecasting & Demand Intelligence Dashboard

Executive Business Report

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Technology Stack: Python, Pandas, NumPy, Scikit-Learn, Statsmodels (SARIMA), Prophet, XGBoost, Streamlit, Plotly, Matplotlib, Git & GitHub

Executive Summary

This project develops an end-to-end Sales Forecasting and Demand Intelligence System using the **Superstore Sales** dataset as the primary source and the **Video Game Sales** dataset as a supplementary dataset for multi-source exploratory analysis. The solution integrates data preprocessing, exploratory data analysis, time-series forecasting, anomaly detection, product demand clustering, customer segmentation, and an interactive Streamlit dashboard. The objective is to support inventory planning, demand forecasting, and executive decision-making using reliable data-driven insights.

Project Objectives

- Data cleaning and feature engineering.
- Historical sales trend and seasonality analysis.
- Forecasting using SARIMA, Prophet and XGBoost.
- Model comparison using MAE, RMSE and MAPE.
- Isolation Forest and Z-Score anomaly detection.
- Product clustering and customer segmentation using K-Means.
- Deployment of an interactive Streamlit dashboard.

Key Business Findings

Technology generated the highest sales contribution while the West and East regions consistently showed strong performance. Sales exhibited a long-term upward trend with recurring seasonality. Average shipping time remained stable across regions, indicating efficient fulfilment operations.

Forecast Summary

SARIMA delivered the best overall forecasting performance and was selected as the recommended production model. The three-month forecast indicates continued growth, especially for the Technology category, while Furniture and Office Supplies are expected to maintain stable demand.

Model Performance Comparison

Model	MAE	RMSE	MAPE
SARIMA	19244.49	19950.07	20.53%
Prophet	20250.79	22318.41	21.86%
XGBoost	36206.50	41996.14	36.71%

Anomaly Detection

Isolation Forest detected 11 anomalous weeks while the Z-Score method identified 2 extreme outliers. These anomalies are likely associated with festive demand, promotional campaigns, or bulk purchasing behaviour.

Segmentation

K-Means clustering grouped products into High, Moderate and Low Demand segments. RFM analysis categorized customers into Premium, Regular and Low Value groups for targeted engagement.

Business Recommendations

1. Prioritize inventory for Technology products.
2. Increase stock before seasonal peaks.
3. Monitor anomalies continuously.
4. Reward premium customers through loyalty programs.
5. Retrain forecasting models periodically.
6. Incorporate holidays, promotions and macroeconomic indicators into future forecasting models.

Business Value

The system improves demand planning, inventory optimization, customer retention, anomaly detection and executive decision-making through an interactive dashboard.

Limitations

Forecasts depend on historical patterns. Unexpected market events, pricing changes or supply-chain disruptions may reduce prediction accuracy.

Project Resources

GitHub: https://github.com/CGNA0524/SalesForecasting_Chirag

Live Dashboard:

<https://salesforecastingchirag-zeugqndw2cjgkwc75qvrfa.streamlit.app/>

Conclusion

This solution demonstrates an industry-standard analytics workflow from raw data to deployment and provides actionable insights for retail demand planning and inventory management.